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Exploring Non-Negative Matrix Factorization: Algorithms and Applications

Samuele Serri

Dipartimento di Matematica Informatica e Geoscienze – MIGE
Università degli studi di Trieste

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Tesi di laurea triennale

Outline

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   - Applications
2. Algorithms
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3. Numerical experiments

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Definition: (NMF problem)

Given a non-negative matrix $V \in \mathbb{R}_+^{m \times n}$ and a factorization rank $r \in \mathbb{N}$, find two non-negative matrices $W \in \mathbb{R}_+^{m \times r}$ and $H \in \mathbb{R}_+^{r \times n}$ that minimize a given cost function $D(V, WH)$, that is:

$$\min_{W \in \mathbb{R}^{m \times r}, H \in \mathbb{R}^{r \times n}} D(V, WH) \tag{1}$$

subject to:

$$W \geq 0, H \geq 0$$

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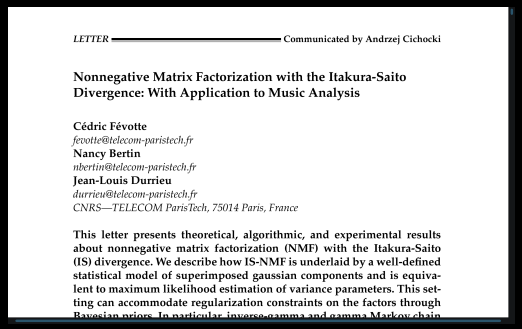
NMF is extremely useful for inherently non-negative datasets
For instance, when the data represent physical quantities like signal strengths or pixel intensities.

Example applications include:

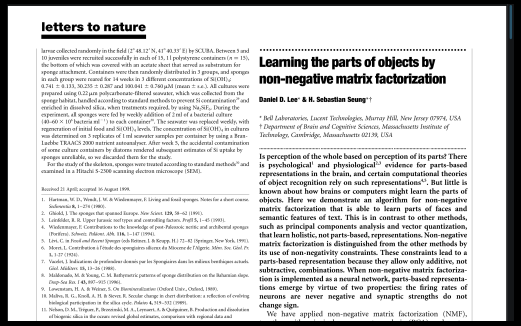
- Audio signal processing.
- Feature extraction in images.
- Topic modeling in text mining.


LETTER Communicated by Andrzej Cichocki

Nonnegative Matrix Factorization with the Itakura-Saito Divergence: With Application to Music Analysis

Cédric Févotte
fevotte@telecom-paristech.fr
Nancy Bertin
nbertin@telecom-paristech.fr
Jean-Louis Durrieu
durrieu@telecom-paristech.fr
CNRS—TELECOM ParisTech, 75014 Paris, France

This letter presents theoretical, algorithmic, and experimental results about nonnegative matrix factorization (NMF) with the Itakura-Saito (IS) divergence. We describe how IS-NMF is underlaid by a well-defined statistical model of superimposed gaussian components and is equivalent to maximum likelihood estimation of variance parameters. This setting can accommodate regularization constraints on the factors through Bayesian priors. In particular, inverse-gamma and gamma Markov chain


letters to nature

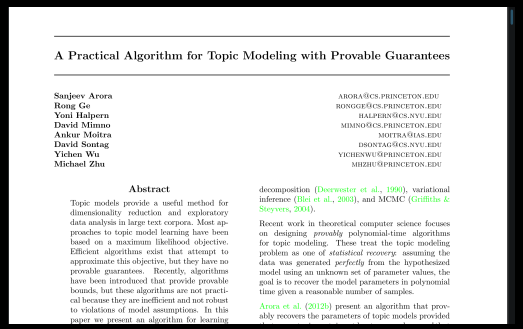
Learning the parts of objects by non-negative matrix factorization

Daniel D. Lee* & H. Sebastian Seung*†

* Bell Laboratories, Lucent Technologies, Murray Hill, New Jersey 07974, USA
† Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, Massachusetts 021.39, USA

Is perception of the whole based on perception of its parts? There is psychological[1] and physiological[2,3] evidence for parts-based representations in the brain, and certain computational theories of object recognition rely on such representations[4,5]. But little is known about how brains or computers might learn the parts of objects. Here we demonstrate an algorithm for non-negative matrix factorization that is able to learn parts of faces and semantic features of text. This is in contrast to other methods, such as principal components analysis and vector quantization, that learn holistic, not parts-based, representations. Non-negative matrix factorization is distinguished from the other methods by its use of non-negativity constraints. These constraints lead to a parts-based representation because they allow only additive, not subtractive, combinations. When non-negative matrix factorization is implemented as a neural network, parts-based representations emerge by virtue of two properties: the firing rates of neurons are never negative and synaptic strengths do not change sign.

We have applied non-negative matrix factorization (NMF),


A Practical Algorithm for Topic Modeling with Provable Guarantees

Sanjeev Arora ARORA@CS.PRINCETON.EDU
Rong Ge RONGGE@CS.PRINCETON.EDU
Yoni Halpern HALPERN@CS.NYU.EDU
David Mimno MIMNO@CS.PRINCETON.EDU
Ankur Moitra MOITRA@IAS.EDU
David Sontag DSONTAG@CS.NYU.EDU
Yichen Wu YICHENWU@PRINCETON.EDU
Michael Zhu MHZHU@PRINCETON.EDU

Abstract

Topic models provide a useful method for dimensionality reduction and exploratory data analysis in large text corpora. Most approaches to topic model learning have been based on a maximum likelihood objective. Efficient algorithms exist that attempt to approximate this objective, but they have no provable guarantees. Recently, algorithms have been introduced that provide provable bounds, but these algorithms are not practical because they are inefficient and not robust to violations of model assumptions. In this paper we present an algorithm for learning

decomposition (Deerwester et al., 1990), variational inference (Blei et al., 2003), and MCMC (Griffiths & Steyvers, 2004).

Recent work in theoretical computer science focuses on designing *provably* polynomial-time algorithms for topic modeling. These treat the topic modeling problem as one of *statistical recovery*: assuming the data was generated *perfectly* from the hypothesized model using an unknown set of parameter values, the goal is to recover the model parameters in polynomial time given a reasonable number of samples.

Arora et al. (2012b) present an algorithm that provably recovers the parameters of topic models provided

Example: learning parts of objects with NMF [2]

Figure: Sample images from the CBCL face dataset.

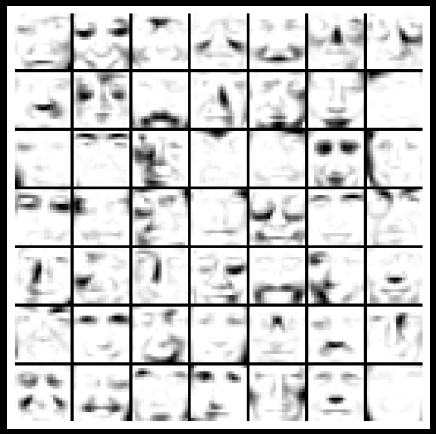


Figure: Basis images obtained from NMF.

- CBCL face dataset $\rightarrow$ 2429 images of faces with 19x19 pixels.
- NMF with $r = 49 \rightarrow 49$ basis images and corresponding coefficients.
- Basis images extract facial features.
- Columns of H contain the weight of each feature for each image.
- Original images reconstructed as $V[:, i] \approx \sum_{k=1}^{r} W[:, k]H[k, i]$.

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Can be interpreted as a scaled gradient descent method:
the gradient of the cost function with respect to H is given by:

$$\nabla_H \frac{1}{2}\|V - WH\|_F^2 = W^T WH - W^T V \tag{2}$$

taking a step of size:

$$\eta_k = \frac{H}{W^T WH} \tag{3}$$

yields the multiplicative update rule:

MU for H

$$H \leftarrow H \circ \frac{W^T V}{W^T WH} \tag{4}$$

MU

Algorithm 1 Multiplicative Update for NMF

Require: $V \in \mathbb{R}_+^{m \times n}$, rank r, initial factors $W^{(0)} \in \mathbb{R}_+^{m \times r}$, $H^{(0)} \in \mathbb{R}_+^{r \times n}$

while stopping criterion not met **do**

$H^{(k+1)} \leftarrow H^{(k)} \circ \frac{W^{(k)\top} V}{W^{(k)\top} W^{(k)} H^{(k)}}$ ▷ Update for H

$W^{(k+1)} \leftarrow W^{(k)} \circ \frac{V H^{(k+1)\top}}{W^{(k)} H^{(k+1)} H^{(k+1)\top}}$ ▷ Update for W

end while

return W, H

PGD

Algorithm 2 Projected Gradient Descent for NMF

Require: $V \in \mathbb{R}_+^{m \times n}$, rank r, initial factors $W^{(0)} \in \mathbb{R}_+^{m \times r}$, $H^{(0)} \in \mathbb{R}_+^{r \times n}$

while stopping criterion not met **do**

  $\mathrm{LW} \leftarrow \|W^{(k)}\|_2^2$

  $\mathrm{LH} \leftarrow \|H^{(k)}\|_2^2$ ▷ Lipschitz constants

  $\mathrm{Grad_W} \leftarrow W^{(k)} H^{(k)} H^{(k)\top} - V H^{(k)\top}$ ▷ Gradient w.r.t. W

  $W^{(k+1)} \leftarrow \max\left(0, W^{(k)} - \frac{1}{LW}\mathrm{Grad_W}\right)$ ▷ Update for W

  $\mathrm{Grad_H} \leftarrow W^{(k+1)\top} W^{(k+1)} H^{(k)} - W^{(k+1)\top} V$ ▷ Gradient w.r.t. H

  $H^{(k+1)} \leftarrow \max\left(0, H^{(k)} - \frac{1}{LH}\mathrm{Grad_H}\right)$ ▷ Update for H

end while

return W, H

ALS

Algorithm 3 Alternating Least Squares for NMF

Require: $V \in \mathbb{R}_+^{m \times n}$, rank r, initial factors $W^{(0)} \in \mathbb{R}_+^{m \times r}$, $H^{(0)} \in \mathbb{R}_+^{r \times n}$

```
while stopping criterion not met do
    for i = 1 to m do
        W(i,:) ← max(0, arg min_w ||V(i,:) − wH^(k)||_2^2)
    end for
    for j = 1 to n do
        H(:,j) ← max(0, arg min_h ||V(:,j) − W^(k)h||_2^2)
    end for
end while
return W, H
```

HALS

Algorithm 4 Hierarchical Alternating Least Squares for NMF

Require: $V \in \mathbb{R}_+^{m \times n}$, rank r, initial factors $W^{(0)} \in \mathbb{R}_+^{m \times r}$, $H^{(0)} \in \mathbb{R}_+^{r \times n}$

while stopping criterion not met **do**
  for $\ell = 1$ to r **do**
    $\tilde{R}_\ell \leftarrow V - \sum_{k \neq \ell} W[:,k]H[k,:]$
    $H[\ell,:] \leftarrow \max\left(0, \frac{\tilde{R}_\ell^\top W[:,\ell]}{W[:,\ell]^\top W[:,\ell]}\right)$
  end for
  for $\ell = 1$ to r **do**
    $\tilde{R}_\ell \leftarrow V - \sum_{k \neq \ell} W[:,k]H[k,:]$
    $W[:,\ell] \leftarrow \max\left(0, \frac{\tilde{R}_\ell H[\ell,:]^\top}{H[\ell,:]H[\ell,:]^\top}\right)$
  end for
end while
return W, H

The Algorithms have been implemenetd in Python

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src	fix random state	4 days ago
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LICENSE	update readme	last month
README.md	update readme	2 minutes ago
pyproject.toml	add toml file	4 months ago

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NMF (Non-negative Matrix Factorization) — Code

The NMF problem consists in: given a non-negative matrix $V \in R^{m \times n}$ and a factorization rank r, find to non-negative matrices $W \in R^{m \times r}$ and $W \in R^{r \times n}$ such that their product approximates the original matrix V as closely as possible.
This is defined in terms of a constrained optimization problem:

$$\min_{W \in R^{m \times r}, H \in R^{r \times n}} D(V, WH) \text{ subject to } W \geq 0, H \geq 0$$

and it is a widely used technics in data analysis whenever dealing with inherently non-negative data.

About

This code was developed as part of my bachelor thesis work at Trieste University. It contains the implementation of algorithms for solving the non-negative matrix factorization (NMF) problem.

algorithms | numerical-methods | non-negative-matrix-factorization

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The Algorithms have been implemenetd in Python

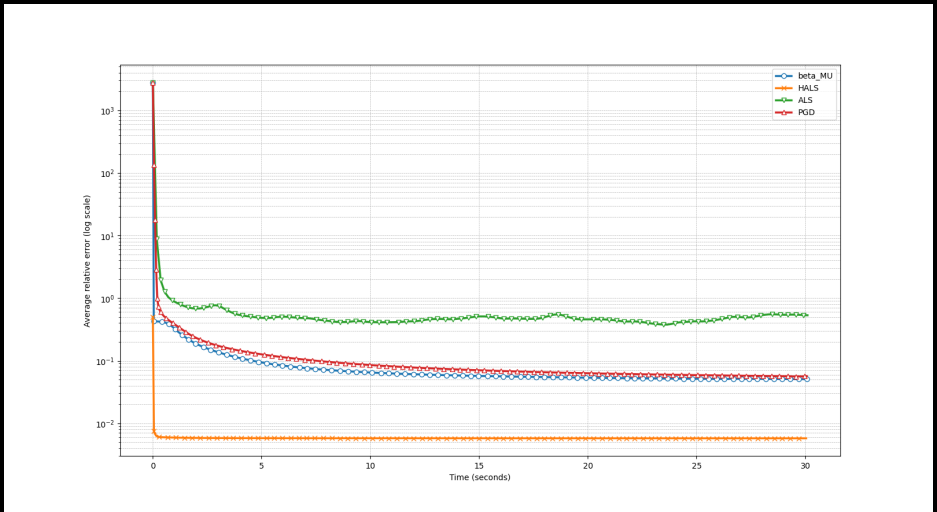

Figure: CBCL dataset:

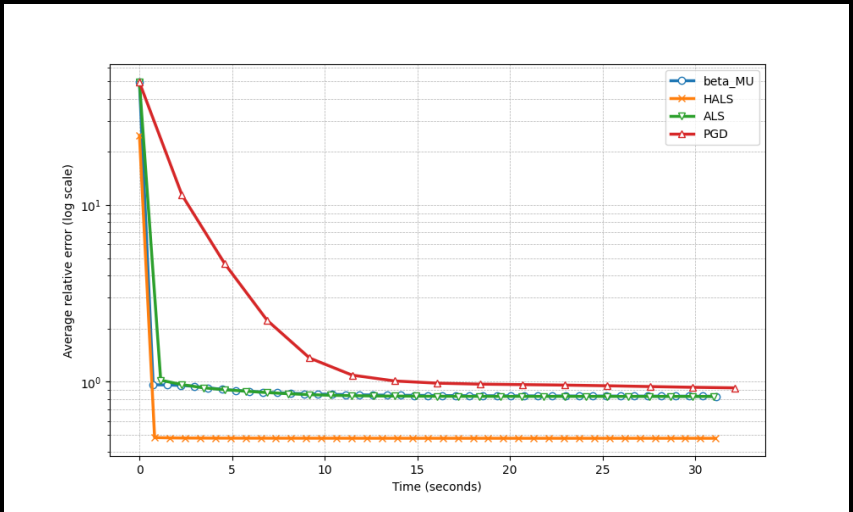

Figure: TDT2 dataset:

Dense case: fastest HALS SParse case: fastest MU

For Further Reading and References I

Gillis, Nicolas.
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SIAM, 2020.

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Nature, 1999.

Thank you for your attention!